StyleCart – Fashion E-Commerce Database Management System

Topic: Database Relationship Analysis using Joins

Objectives

1. Combine Customer, Product, Order, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

1. INNER JOIN — matching records only

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  INNER JOIN Orders o
  9      ON c.Customer_ID = o.Customer_ID
 10  ORDER BY c.Customer_ID;

Customer ID First Name        ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------- --------------- ---------- --------- ------------
          1 Arun                  1001 20-OCT-26         1898
          1 Arun                  1003 22-OCT-26          899
          2 Priya                 1002 21-OCT-26         1599
          3 Rahul                 1004 23-OCT-26          999
          5 Karthik               1005 24-OCT-26          599
```

2. LEFT JOIN — all customers

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      o.Order_ID,
  5      o.Total_Amount
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8      ON c.Customer_ID = o.Customer_ID
  9  ORDER BY c.Customer_ID;

Customer ID First Name        ORDER_ID TOTAL_AMOUNT
----------- --------------- ---------- ------------
          1 Arun                  1001         1898
          1 Arun                  1003          899
          2 Priya                 1002         1599
          3 Rahul                 1004          999
          4 Divya
          5 Karthik               1005          599
```

3. RIGHT JOIN — All orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      o.Order_ID,
  5      o.Total_Amount
  6  FROM Customer c
  7  RIGHT JOIN Orders o
  8      ON c.Customer_ID = o.Customer_ID
  9  ORDER BY o.Order_ID;

Customer ID First Name        ORDER_ID TOTAL_AMOUNT
----------- --------------- ---------- ------------
          1 Arun                  1001         1898
          2 Priya                 1002         1599
          1 Arun                  1003          899
          3 Rahul                 1004          999
          5 Karthik               1005          599
```

4.Retrieve complete order details.

```
Cust ID First Name   Last Name    Order ID Order Date Product         Qty Unit Price Subtotal Order Total Order Stat P
ayment Method   Payment Stat Amount Paid
------- ------------ ------------ -------- ---------- --------------- --- ---------- -------- ----------- ---------- -
----------------- ------------ -----------
      1 Arun         Kumar           1001 20-OCT-26  T-Shirt           1        599      599        1898 Delivered  U
PI              Successful          1898
      1 Arun         Kumar           1001 20-OCT-26  Jeans             1       1299     1299        1898 Delivered  U
PI              Successful          1898
      2 Priya        Sharma          1002 21-OCT-26  Sports Shoes      1       1599     1599        1599 Shipped    C
redit Card      Successful          1599
      1 Arun         Kumar           1003 22-OCT-26  Kurti             1        899      899         899 Placed     D
ebit Card       Failed               899
      3 Rahul        Raj             1004 23-OCT-26  Formal Shirt      1        999      999         999 Delivered  U
PI              Successful           999
      5 Karthik      Mohan           1005 24-OCT-26  T-Shirt           1        599      599         599 Placed     C
ash on Delivery Pending              599

6 rows selected.
```

5.Display customer purchase history

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      o.Order_ID,
  6      o.Order_Date,
  7      pr.Product_Name,
  8      oi.Quantity,
  9      oi.Unit_Price,
 10      o.Order_Status
 11  FROM Customer c
 12  INNER JOIN Orders o
 13      ON c.Customer_ID = o.Customer_ID
 14  INNER JOIN Order_Item oi
 15      ON o.Order_ID = oi.Order_ID
 16  INNER JOIN Product pr
 17      ON oi.Product_ID = pr.Product_ID
 18  ORDER BY c.Customer_ID, o.Order_Date;

Cust ID First Name   Last Name    Order ID Order Date Product         Qty Unit Price Order Stat
------- ------------ ------------ -------- ---------- --------------- --- ---------- ----------
      1 Arun         Kumar           1001 20-OCT-26  T-Shirt           1        599 Delivered
      1 Arun         Kumar           1001 20-OCT-26  Jeans             1       1299 Delivered
      1 Arun         Kumar           1003 22-OCT-26  Kurti             1        899 Placed
      2 Priya        Sharma          1002 21-OCT-26  Sports Shoes      1       1599 Shipped
      3 Rahul        Raj             1004 23-OCT-26  Formal Shirt      1        999 Delivered
      5 Karthik      Mohan           1005 24-OCT-26  T-Shirt           1        599 Placed

6 rows selected.
```

5.1 To view only one customer's purchase history

```
SQL> SELECT
  2      c.First_Name,
  3      c.Last_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      pr.Product_Name,
  7      oi.Quantity,
  8      oi.Unit_Price,
  9      o.Order_Status
 10  FROM Customer c
 11  JOIN Orders o
 12      ON c.Customer_ID = o.Customer_ID
 13  JOIN Order_Item oi
 14      ON o.Order_ID = oi.Order_ID
 15  JOIN Product pr
 16      ON oi.Product_ID = pr.Product_ID
 17  WHERE c.Customer_ID = 1
 18  ORDER BY o.Order_Date;

First Name   Last Name    Order ID Order Date Product         Qty Unit Price Order Stat
------------ ------------ -------- ---------- --------------- --- ---------- ----------
Arun         Kumar            1001 20-OCT-26  T-Shirt           1        599 Delivered
Arun         Kumar            1001 20-OCT-26  Jeans             1       1299 Delivered
Arun         Kumar            1003 22-OCT-26  Kurti             1        899 Placed
```

6. Generate multi-table reports.

6.1Report 1: Customer-wise order summary

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      COUNT(o.Order_ID) AS Total_Orders,
  6      NVL(SUM(o.Total_Amount), 0) AS Total_Purchase
  7  FROM Customer c
  8  LEFT JOIN Orders o
  9      ON c.Customer_ID = o.Customer_ID
 10  GROUP BY
 11      c.Customer_ID,
 12      c.First_Name,
 13      c.Last_Name
 14  ORDER BY c.Customer_ID;

Cust ID First Name   Last Name    TOTAL_ORDERS TOTAL_PURCHASE
------- ------------ ------------ ------------ --------------
      1 Arun         Kumar                   2           2797
      2 Priya        Sharma                  1           1599
      3 Rahul        Raj                     1            999
      4 Divya        Rani                    0              0
      5 Karthik      Mohan                   1            599
```

Report 2: Product-wise sales report

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      NVL(SUM(oi.Quantity), 0) AS Quantity_Sold,
  5      NVL(SUM(oi.Quantity * oi.Unit_Price), 0) AS Total_Sales
  6  FROM Product p
  7  LEFT JOIN Order_Item oi
  8      ON p.Product_ID = oi.Product_ID
  9  GROUP BY
 10      p.Product_ID,
 11      p.Product_Name
 12  ORDER BY Total_Sales DESC;

PRODUCT_ID Product         QUANTITY_SOLD TOTAL_SALES
---------- --------------- ------------- -----------
       104 Sports Shoes                1        1599
       102 Jeans                       1        1299
       101 T-Shirt                     2        1198
       105 Formal Shirt                1         999
       103 Kurti                       1         899
```

Report 3: Payment status report

```
SQL> SELECT
  2      o.Order_ID,
  3      c.First_Name,
  4      o.Total_Amount,
  5      p.Payment_Method,
  6      p.Payment_Status,
  7      p.Amount_Paid
  8  FROM Orders o
  9  JOIN Customer c
 10      ON o.Customer_ID = c.Customer_ID
 11  LEFT JOIN Payment p
 12      ON o.Order_ID = p.Order_ID
 13  ORDER BY o.Order_ID;

Order ID First Name   Order Total Payment Method     Payment Stat Amount Paid
-------- ------------ ----------- ------------------ ------------ -----------
    1001 Arun                1898 UPI                Successful          1898
    1002 Priya               1599 Credit Card        Successful          1599
    1003 Arun                 899 Debit Card         Failed               899
    1004 Rahul                999 UPI                Successful           999
    1005 Karthik              599 Cash on Delivery   Pending              599
```